

BAHA EDDINE TAYARI

Electrical & Automation Engineering Student

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🌐 [LinkedIn](#) | 🌐 [Portfolio](#)

Profile

Final-year Electrical & Automation Engineering student at ENIG focused on **embedded systems, industrial automation and intelligent cyber-physical systems**. Hands-on with **C/C++, ESP32, Arduino, Siemens S7/TIA Portal and industrial HMI**. Seeking a **5–6 month PFE internship from February 2027** in embedded systems, automation, Edge AI or Industry 4.0.

Education

State Engineering Degree – Electrical Engineering & Automation Sep. 2024 – Present
École Nationale d'Ingénieurs de Gabès (ENIG)

Preparatory Cycle – Mathematics & Physics (MP) Sep. 2021 – Jun. 2024
Institut Préparatoire aux Études d'Ingénieurs de Bizerte (IPEIB) Baccalauréat Mathematics, Jul. 2021

Engineering Experience

Safran Electronics & Defense Jun. – Aug. 2026
Engineering Intern – Methods / Industrial Engineering Tunisia

- Designed and prototyped an automated rodage test system with **Arduino Mega, motor control, timing, display and mechanical tooling** to improve repeatability.
- Analyzed aerospace manufacturing constraints with operators and methods engineers; improved workstation ergonomics and process quality using **SolidWorks, Proteus and Arduino IDE**.

Tunisie Câbles – OneTech Group Jul. – Aug. 2025
Automation Intern Tunisia

- Worked with **Siemens S7-1200 PLCs, TIA Portal, WinCC HMI, VFDs and instrumentation** in cable manufacturing.
- Investigated maintenance/automation issues involving temperature, voltage drops, motors, relays and fuses; studied an automatic granulate-loading system with PLC and WinCC supervision.

Selected Technical Projects

- STM32F103 Bare-Metal Acquisition System** – low-level drivers, timers, interrupts, UART/SPI/I2C without HAL. [prog C](#) / [STM32](#) / [ARM](#)
- ESP32 LoRa Predictive Maintenance Platform** – IoT monitoring of temperature/current/vibration with Streamlit dashboard. [C++](#) / [ESP32](#) / [LoRa](#) / [Python](#)
- Automated Rodage Test System** – motor actuation, timing, display and mechanical tooling for production testing. [Arduino](#) / [Proteus](#) / [SolidWorks](#)
- Smart House Monitoring Interface** – HMI-style app with real-time visualization, database and multilingual support. [Python](#) / [Streamlit](#) / [SQLite](#)

Technical Skills

Embedded	C/C++, STM32F103, ESP32, Arduino, bare-metal, timers, interrupts, UART, SPI, I2C, CAN, RS485
Automation	Siemens S7-1200/1500, TIA Portal, Ladder, WinCC, PLC, HMI, Modbus, industrial instrumentation
AI / Data	Python, ML fundamentals, Edge AI, Streamlit, SQLite, Plotly
Tools	Proteus ISIS, STM32CubeIDE, Arduino IDE, SolidWorks, MATLAB, PVsyst, Git
Systems	PWM, motor control, power electronics fundamentals, TCP/IP, LoRa, embedded communication
Certifications	

Machine Learning at the Edge on Arm – Arm (Jun. 2026) | **EF SET English** – 84/100, C2 (Aug. 2026) | **Cisco**: Python Essentials, Introduction to IoT, Modern AI, Data Analytics (2025)

Languages & Additional Experience

Languages: English – C2 | French – Professional | Arabic – Native
Additional Experience: ISIE Tunisia – Election Operations & Coordination | Agricultural Field Operations | Sales & Customer Relations